Linear Graphs 4: Drawing Lines

Name _____ **Date** _____ No calculator Version 1

KEY WORDS table of values · plot · gradient · y-intercept · rearrange · horizontal · vertical · intercepts

A By a table of values ●○○

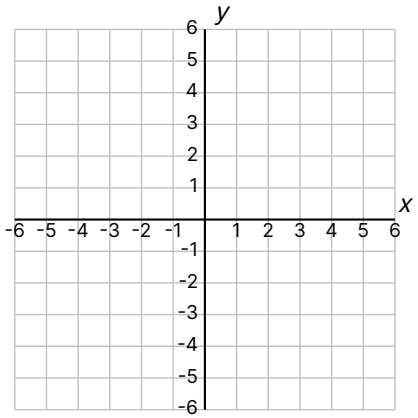
ANSWER ON THE SHEET

Remember: Work out y for each x in the table, plot the points, then rule a straight line through them and past them to the edge of the grid.

1. Complete each table, then draw the line on the grid.

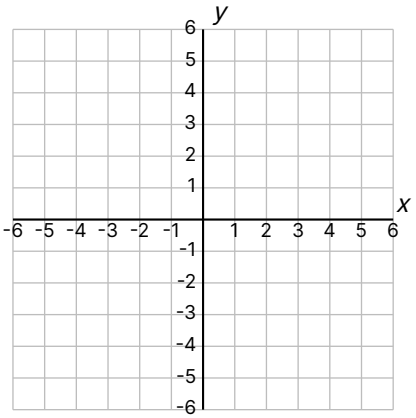
a) $y = x + 2$

x	-3	0	3
y			



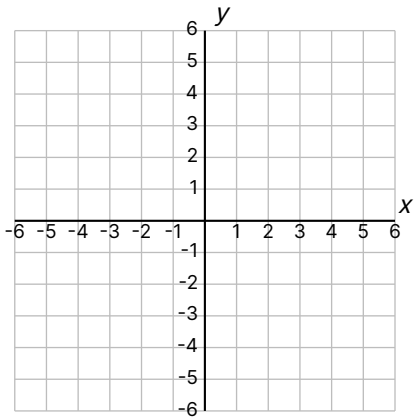
b) $y = 2x - 3$

x	-1	0	2
y			



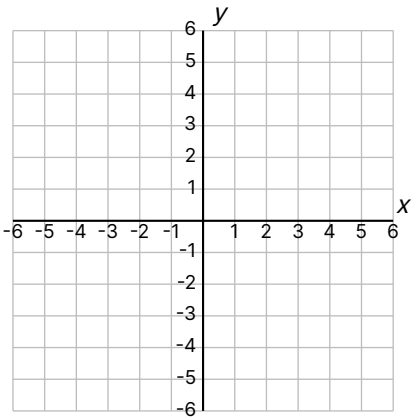
c) $y = -x + 1$

x	-2	0	4
y			



d) $y = 4 - 2x$

x	0	2	4
y			



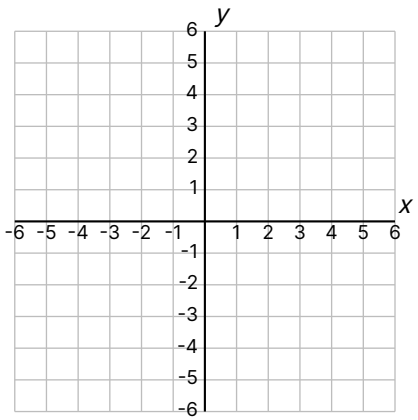
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B From the gradient and intercept ●●○ ANSWER ON THE SHEET

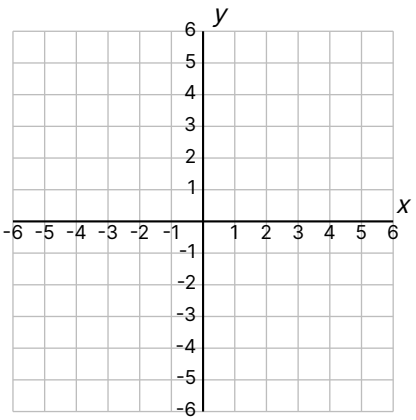
Remember: No table needed: plot the y-intercept, then step along by the gradient (for $m = 2$, go across 1 and up 2) to find more points. Rule the line and label it with its letter.

2. Draw the three lines on each grid. Label each line with its letter.

a) A: $y = 2x + 1$, B: $y = -x + 3$, C: $y = \frac{1}{2}x - 2$



b) A: $y = -3x + 2$, B: $y = x - 4$, C: $y = -2x + 3$

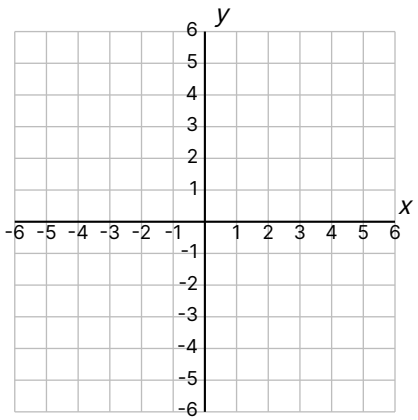


C Horizontal, vertical and through the origin ●●○ ANSWER ON THE SHEET

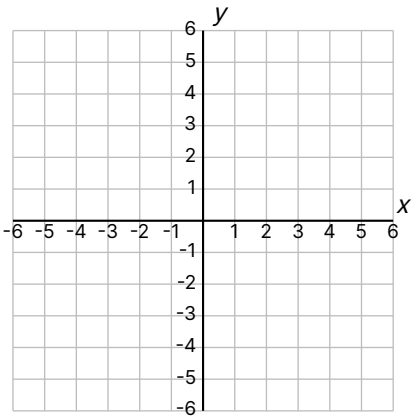
Remember: $y = 3$ is a horizontal line through 3 on the y-axis. $x = -2$ is a vertical line through -2 on the x-axis. A line with no c goes through the origin.

3. Draw the four lines on each grid. Label each line with its letter.

a) A: $y = 3$, B: $x = -2$, C: $y = 2x$, D: $y = -x$



b) A: $y = -4$, B: $x = 4$, C: $y = \frac{1}{2}x$, D: $y = -3x$



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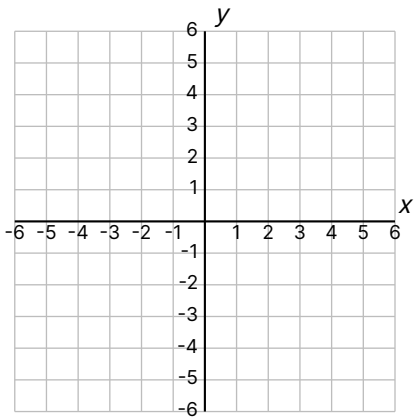
D Fractional gradients ●●●

ANSWER ON THE SHEET

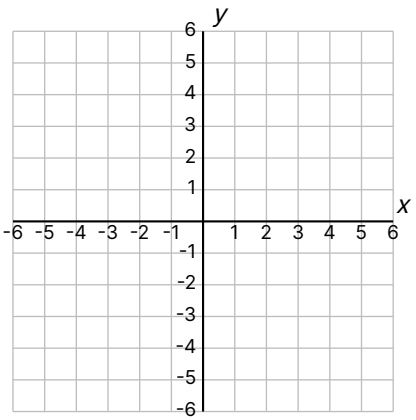
Remember: For $m = \frac{2}{3}$, go across 3 and up 2. For $m = -\frac{3}{2}$, go across 2 and down 3. Step both ways from the intercept so the line is long enough to rule accurately.

4. Draw the three lines on each grid. Label each line with its letter.

a) A: $y = \frac{2}{3}x + 1$, B: $y = -\frac{3}{2}x - 1$, C: $y = -\frac{1}{2}x + 4$



b) A: $y = \frac{3}{4}x - 3$, B: $y = -\frac{2}{3}x + 2$, C: $y = \frac{1}{3}x + 1$



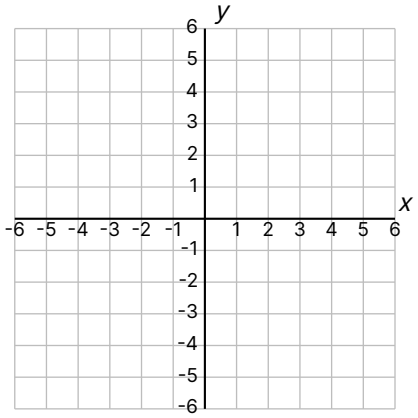
E Equations in other forms ●●●

ANSWER ON THE SHEET

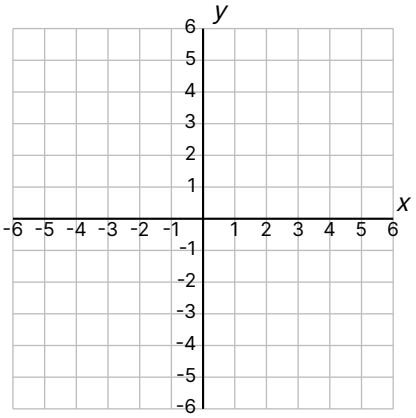
Remember: Rearrange into $y = mx + c$ first: $2y = 4x - 2$ becomes $y = 2x - 1$. For $x + y = 4$ you can instead find the two intercepts: $x = 0$ gives $y = 4$, and $y = 0$ gives $x = 4$.

5. Draw the three lines on each grid. Label each line with its letter.

a) A: $2y = 4x - 2$, B: $x + y = 4$, C: $3x - 2y = 6$



b) A: $2y = x + 4$, B: $2x + y = 5$, C: $2x + 3y = -3$

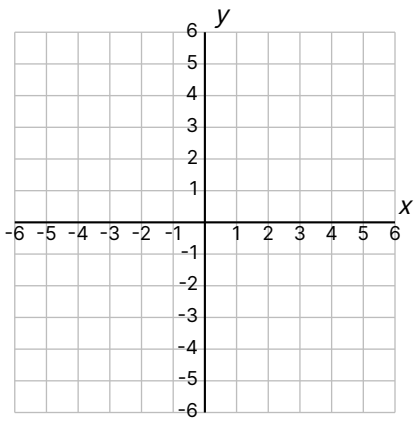


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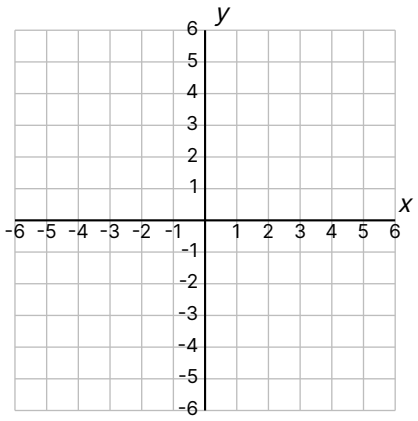
Extension — FOR EARLY FINISHERS

6. Draw each line on the grid, then answer in your book.

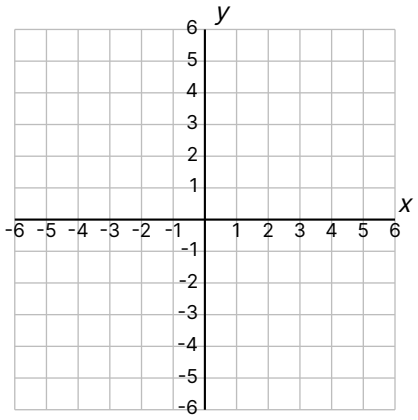
a) A: $y = x - 2$ and B: $y = -2x + 4$. Write down the point where they cross.



b) The line through $(1,\ 2)$ with gradient -2. Then write its equation.



c) The line with x-intercept 4 and y-intercept -2. Then write its equation.



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Linear Graphs 4: Drawing Lines

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS table of values · plot · gradient · y-intercept · rearrange · horizontal · vertical · intercepts

A By a table of values ●○○

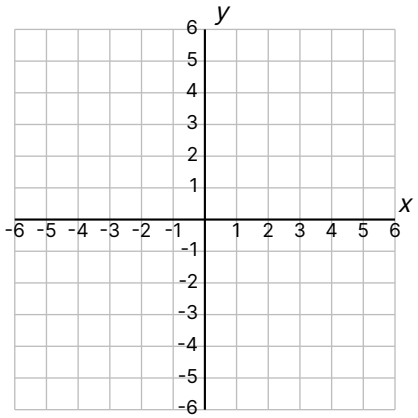
ANSWER ON THE SHEET

Remember: Work out y for each x in the table, plot the points, then rule a straight line through them and past them to the edge of the grid.

1. Complete each table, then draw the line on the grid.

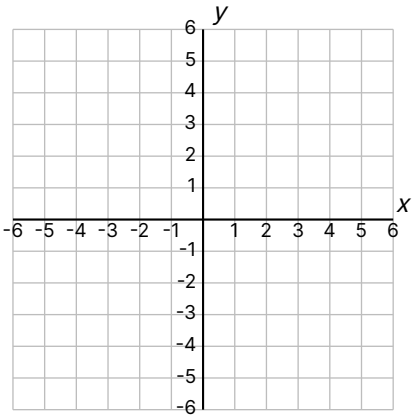
a) $y = x - 2$

x	-3	0	3
y			



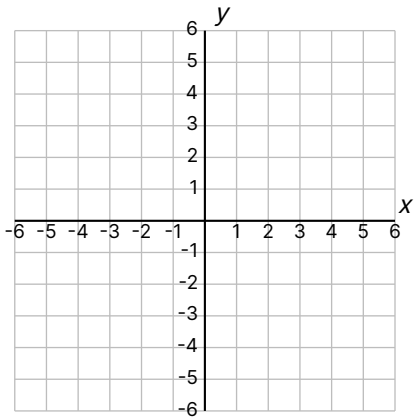
b) $y = 2x + 1$

x	-2	0	2
y			



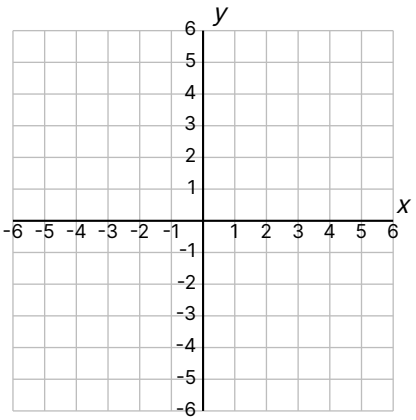
c) $y = -x + 3$

x	-2	0	4
y			



d) $y = 5 - 2x$

x	0	2	4
y			



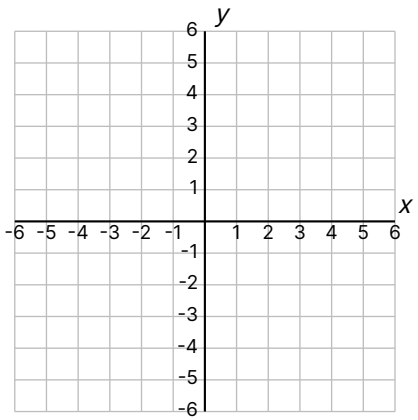
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B From the gradient and intercept ●●○ ANSWER ON THE SHEET

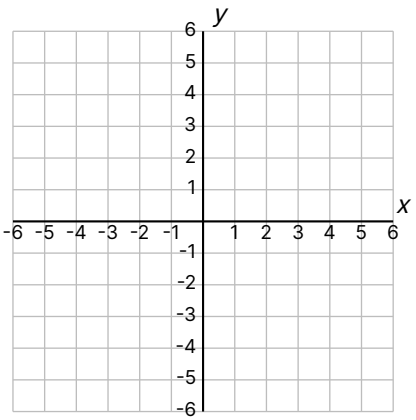
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2. Draw the three lines on each grid. Label each line with its letter.

a) A: $y = 2x - 1$, B: $y = -x + 2$, C: $y = \frac{1}{2}x + 1$



b) A: $y = -3x + 4$, B: $y = x + 3$, C: $y = -2x - 3$

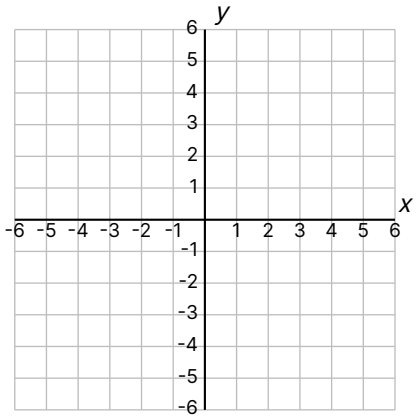


C Horizontal, vertical and through the origin ●●○ ANSWER ON THE SHEET

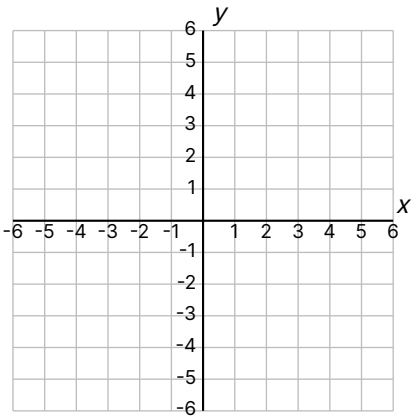
Remember: $y = 3$ is a horizontal line through 3 on the y-axis. $x = -2$ is a vertical line through -2 on the x-axis. A line with no c goes through the origin.

3. Draw the four lines on each grid. Label each line with its letter.

a) A: $y = -3$, B: $x = 4$, C: $y = 3x$, D: $y = -x$



b) A: $y = 4$, B: $x = -4$, C: $y = -\frac{1}{2}x$, D: $y = 2x$



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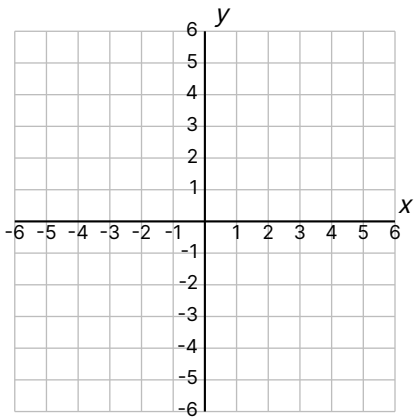
D Fractional gradients ●●●

ANSWER ON THE SHEET

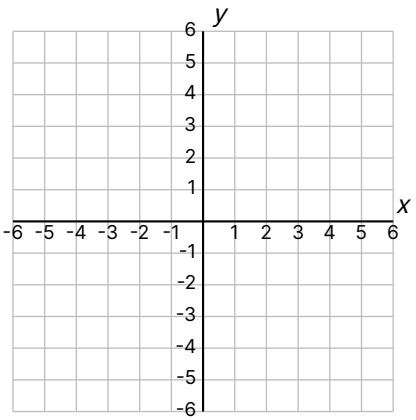
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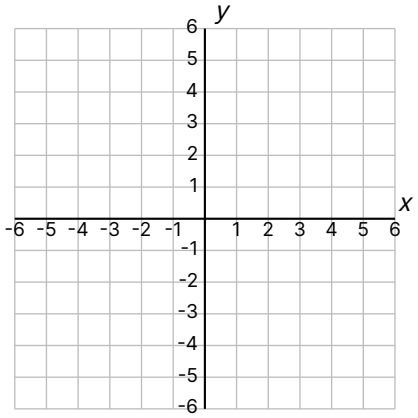
E Equations in other forms ●●●

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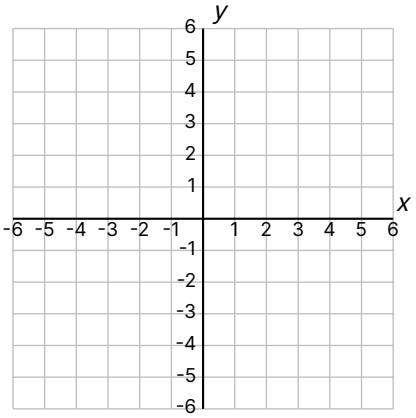
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5. Draw the three lines on each grid. Label each line with its letter.

a) A: $2y = 6x - 4$, B: $x + y = 5$, C: $2x - 3y = 6$



b) A: $2y = x - 2$, B: $3x + y = 4$, C: $3x + 4y = 12$

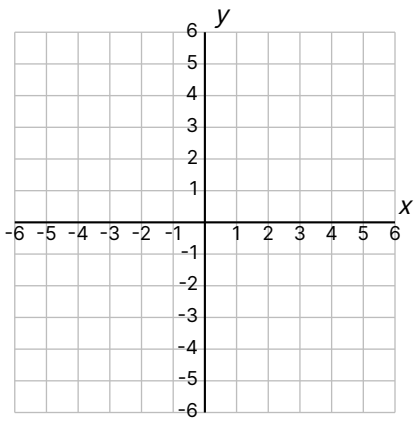


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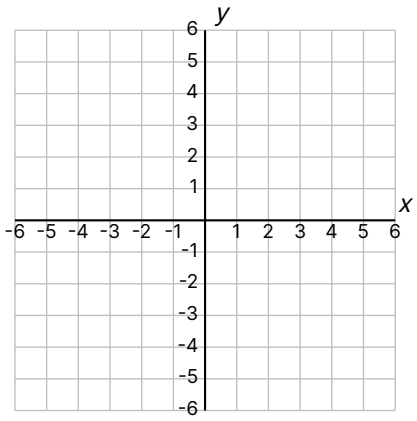
Extension — FOR EARLY FINISHERS

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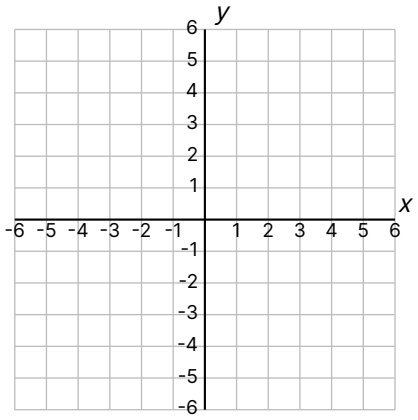
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b) The line through $(-1,\ 3)$ with gradient 2. Then write its equation.



c) The line with x-intercept -3 and y-intercept 2. Then write its equation.



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